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Admissions Committee
MSc Business Analytics
Vrije Universiteit Amsterdam

Dear Members of the Admissions Committee,

My Bachelor's degree in Business Information Systems at the Alexandru Ioan Cuza University of Iaşi sits deliberately at the intersection of two disciplines: rigorous quantitative analysis and applied software engineering. From my first semester, I made a conscious choice to pursue the most technically demanding modules available within my curriculum, driven by a conviction that the most impactful business decisions are those grounded in both computational precision and economic understanding. The MSc in Business Analytics at VU Amsterdam is the most suitable next step for me in that direction - a program that does not treat data science and business strategy as separate fields, but as a single discipline.

I have always been someone who enjoys working with people and being part of decisions - but what my undergraduate studies added was a genuine interest in the analytical side of those decisions. I realised that what I enjoy most is not just communicating conclusions, but building them from data first. The chosen programme felt like the natural meeting point between the two: a program whose combination of **Applied Stochastic Modelling, Optimization of Business Processes, and Advanced Machine Learning** mirrors the analytical framework I have been building throughout my degree. I am also drawn to VU's standing in QS Subject Rankings for Business and Management and to its location within Amsterdam - a hub where financial institutions, technology companies, and consultancies operate in direct proximity to academic research. I intend to take full advantage of that environment.

My 180 ECTS curriculum covers 91 ECTS across Mathematics, Statistics, Econometrics, and Computer Science. The mapping to your program's core modules is direct:

- **Applied Stochastic Modelling & Operational Research:** My courses in Applied Mathematics in Economics (5 ECTS) and Operational Research (5 ECTS) cover Markov chains, the Simplex algorithm, continuous and discrete optimisation, and graph theory - the exact content listed in your Applied Stochastic Modelling and Optimisation module descriptions.
- **Statistical Models & Econometrics:** My Statistics and Probability Theory course (6 ECTS) and Econometrics (5 ECTS) provide a solid foundation in hypothesis testing, regression modelling (linear and non-linear), and inferential statistics, directly aligned with your Statistical Models module.
- **Data Mining, Machine Learning & Programming:** With 24 ECTS in formal programming (Java and C#, OOP paradigm) and 6 ECTS in Database Management (PostgreSQL, advanced SQL), I significantly exceed the minimum programming requirements. My Algorithms and Programming Logic course covered computational complexity analysis, recursion, and formal proof structures — mathematical thinking applied through code.

- **Decision Support & Business Process Design:** My Information Systems Analysis course (covering SDLC, formal process modelling, and UML) combined with a group project designing a decision support system provide practical grounding for your Project Optimisation of Business Processes module. Four Year 3 courses - Integrated Systems (ERP), Information Systems Design, and Information Systems Security, Web Development - are scheduled for completion by July 2026 and will further strengthen this foundation.

I want to address the admission requirement of 45 ECTS in Mathematics directly and transparently.

My curriculum formally categorises 25 ECTS under Mathematics and Statistics. The remaining quantitative content is embedded within Computer Science and Econometrics modules, as is standard for a Business Information Systems degree. I am not claiming that Computer Science is equivalent to pure mathematics. I am making a more specific claim: the courses I completed required, and developed, the same mathematical competencies your program builds upon.

Concretely: my Algorithms course required formal proof by induction and mathematical complexity analysis; my Operational Research course required linear programming and graph optimisation; my Econometrics course required matrix algebra and statistical inference. These are not peripheral skills - they are the direct foundations of stochastic modelling and machine learning. I am prepared to provide detailed syllabi for any course upon request.

My academic performance across all quantitative courses has been consistently strong. I believe that demonstrated results in technically demanding subjects are the most reliable indicator of readiness for graduate-level analytical work, and I am confident my transcript supports that case.

My professional experience has reinforced the analytical skills developed academically. As a Financial Adviser at OVB Allfinanz, I translated complex financial data into personalised investment strategies for clients with diverse risk profiles - a role that demanded both quantitative judgement and clear communication. My QA internship at Pentalog (Globant) placed me in an Agile environment where I worked with Jira, wrote User Stories, and translated functional requirements into structured test coverage. Beyond formal employment, my participation in the CodeLess Lab - a laboratory established in partnership with Codeless Technology B.V. of the Netherlands - gave me direct exposure to the Dutch professional and technological environment I am applying to enter. These experiences confirmed what I intend to build my career on: acting as the bridge between technical rigour and business utility.

I am applying to the MSc Business Analytics at VU Amsterdam as my first and only choice for postgraduate study, because I am already convinced that this is exactly where I want to be. I am fully committed to the demands of the program and to contributing actively to the academic community at VU.

Sincerely,

Sabina Alexia Podina